Exercise – *Down the Rabbit Hole*

"Oh my ears and whiskers, how late it's getting!"

Alice's Adventures in Wonderland *by Lewis Carroll (1832–1898)*

The White Rabbit is anxiously anticipating Alice's visit to Wonderland, worrying that she may have trouble dealing with the challenges it presents. Thus, he wants to equip Alice with a number of magical cakes that may aid her on her adventures. Since Alice is already on her way to the rabbit hole, time is of the essence.

The cakes are stored in various rooms of the White Rabbit's underground home. The home also has a unique *entry* (which also serves as an exit). The rooms are connected by a network of two-way tunnels of various lengths. Each such tunnel connects either two rooms, or it connects a room and the entry. The home is designed such that from every room there is a unique sequence of tunnels that leads to the entry. The White Rabbit starts at the entry and traverses the tunnels so as to visit all rooms of his home and gather as many cakes as possible. To this end, he may traverse each tunnel at most twice (once in either direction), and has to exit back to the surface once he is done collecting the cakes.

However, magical cakes are tricky to deal with, as they have the annoying tendency to magically disappear: For every second that passes after the White Rabbit enters its home, one cake disappears from every room. That is, if the White Rabbit arrives at some room s seconds after entering its home, then s cakes have already disappeared from this room. The White Rabbit never rests and travels at a speed of one meter per second. He has an enchanted bag that instantly collects all cakes left in a room as soon as the rabbit enters it and stores them safely so that no cake can disappear from the bag. The rooms are small and the cakes are handled by the enchanted bag; thus, traversing rooms and collecting cakes takes no time for practical purposes. Furthermore, you may assume that in every room there are sufficiently many cakes such that at least one is left when the White Rabbit enters the room (no matter in what order he decides to visit the rooms).

What is the maximum possible number of magical cakes the White Rabbit can gather?

Input The first line of the input contains the number $t \leqslant 30$ of test cases. Each of the t test cases is described as follows.

- It starts with a line that consists of a single integer n denoting the number of rooms in the home of the White Rabbit ($2 \leqslant n \leqslant 5 \cdot 10^4$).
- The following line consists of n integers $c_1\ c_2\ \ldots\ c_n$, separated by a space, denoting the number of magical cakes stored in the i-th room ($1 \leqslant c_i < 2^{31}$).
- The following n lines define the tunnels. The i-th such line consists of three integers $a\ b\ l$, separated by a space, denoting the two rooms it connects ($0 \leqslant a < b \leqslant n$) and the length of the tunnel in meters ($1 \leqslant \ell \leqslant 2^8$). You may assume that a is *strictly closer to* the entry than b, in terms of the number of tunnels needed to reach the entry. If $a = 0$, the tunnel connects the entry (labeled 0) and the room b.

Output For each test case output a single line with the maximum number of cakes the White Rabbit can gather from the underground network.

Points There are four groups of test sets worth 80 points in total. For each group there is also a corresponding hidden test set that is worth 5 points. So, you can achieve $80 + 4 \cdot 5 = 100$ points overall.

Let c be either the entry or a room and let $f_1, \ldots, f_k$ be the rooms directly connected to e via tunnels so that c is strictly closer to the entry than any f_i. A rabbit home is *uniform* if for every such c the number of rooms reachable from f_i without going through e is *exactly the same for all* $i \in \{1, \ldots, k\}$.

1. For the first group of test sets, worth 25 points, you may assume that $n \leqslant 10^3$, the White Rabbit's home is uniform, the entry point has exactly two tunnels leading to/from it, and every room has at most three tunnels leading to/from it.
2. For the second group of test sets, worth 25 points, you may assume that the White Rabbit's home is uniform, the entry point has exactly two tunnels leading to/from it, and every room has at most three tunnels leading to/from it.
3. For the third group of test sets, worth 25 points, you may assume that the White Rabbit's home is uniform.
4. For the fourth group of test sets, worth 5 points, there are no additional assumptions.

Sample Input

```
2
4
17 15 16 15
0 1 1
0 2 2
1 3 3
2 4 1
5
20 20 20 20 20
0 1 1
0 2 2
2 3 2
2 4 1
3 5 2
```

Sample Output

```
41
72
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